

Sonna's Patisserie and Cafe - Application Development Report

Executive Summary

This report outlines the technological foundations and the step-by-step implementation plan for building the mobile application and backend services for **Sonna's Patisserie and Cafe**. The project is designed with a strong focus on "Editorial Elegance", combining a reliable, typesafe backend with a premium, glassmorphic, and highly interactive user experience on mobile devices. Furthermore, the application is being architected to integrate seamless WhatsApp automation to facilitate organic customer interaction and order management.

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Part 1: Technology Stack & Version Strategy

The application adopts a decoupled architecture with a mobile client built in Flutter and a strictly typed API backend built on Next.js and Prisma. We have chosen specific technologies to guarantee a "luxury" look and feel while keeping operational costs low and developer velocity high.

1.1 Mobile Application (Frontend)

The mobile app handles user interaction, presenting the elegant menu, and facilitating orders.

Technology	Version	Purpose & Rationale
Dart SDK	<code>^3.10.1</code>	The underlying language for Flutter, providing sound null safety and excellent performance.
Flutter	Latest	A cross-platform UI toolkit. We chose Flutter to deliver a highly customized, animation-rich, 60fps luxury experience simultaneously on both iOS and Android from a single codebase.
google_fonts	<code>^8.0.2</code>	Delivers premium typography (<i>Noto Serif</i> and <i>Plus Jakarta Sans</i>). Typography is the most critical element of the editorial luxury aesthetic.

lucide_icons	<code>^0.257.0</code>	Provides a consistent, modern, and clean set of vector icons for the user interface.
fl_chart	<code>^1.2.0</code>	Allows for future expansion into rich data visualizations, such as order statistics tracking for user profiles or admin views.

1.2 Backend API, Database, & Integrations

The backend manages data persistence (cakes, orders, users), authenticates sessions, and connects to third-party services like WhatsApp.

Technology	Version	Purpose & Rationale
Next.js	<code>^15.2.3</code>	Functions as the primary API host. It provides easy serverless deployment, intelligent routing, and an excellent developer experience.
React	<code>^19.0.0</code>	The core engine behind Next.js. While primarily used as an API layer, keeping React enables us to quickly spin up functional web dashboards or admin panels in the future.
TypeScript	<code>^5.8.2</code>	Provides static typing to eliminate broad classes of bugs before code is even run. Essential for a stable enterprise-grade backend.
Prisma ORM	<code>^6.6.0</code>	A modern Database ORM. It enables declarative data modeling, automatic migration generation, and complete end-to-end type safety when interacting with the database.
PostgreSQL	Supabase	A managed cloud database service. Perfect for safely managing transactional data like user accounts and order histories.
tRPC	<code>^11.0.0</code>	Remote Procedure Call architecture. Allows us to declare API endpoints as simple functions and consume them on the frontend with inferred types, completely avoiding REST/GraphQL boilerplate.
NextAuth.js	<code>v5 beta</code>	A comprehensive authentication solution paired with <code>@auth/prisma-adapter</code> (<code>^2.7.2</code>) to effortlessly integrate with our database to manage secure user sessions.
WhatsApp API	-	[Added Requirement] Integration with WhatsApp Cloud API to automate order confirmations, send tracking details, and allow customers to directly ask questions or initiate orders via WhatsApp.

Part 2: Product Catalog & Database Seeding

The following exact products will be seeded into the database as the initial catalog for Sonna's Patisserie and Cafe. All cakes are priced consistently across two tiers:

- **600 grams** (Serves 4-6): 750/- (*Exception: Classic Chocolate is 675/-*)
- **1050 grams** (Serves 8-12): 1350/- (*Exception: Classic Chocolate is 1250/-*)

Category: Chocolate Based

1. SONNA'S CLASSIC CHOCOLATE

- *Description:* Classic chocolate cake with chocolate whipped ganache.
- *Pricing:* 600g (₹675) | 1050g (₹1250)

2. ALMOND BRITTLE WITH SALTED CARAMEL GANACHE

- *Description:* A Sonna's original: In house caramel with caramel chocolate ganache with almond brittle.
- *Tags:* Caramel, Caramel Chocolate Ganache, Almond Brittle, Chocolate cake, Chocolate Whipped Ganache.

3. ORANGE & CHOCOLATE

- *Description:* Classic chocolate cake with orange chocolate whipped ganache.
- *Tags:* Chocolate, Orange.

4. HAZELNUT & CHOCOLATE

- *Description:* Chocolate cake with hazelnut ganache (chopped almonds optional).
- *Tags:* Chocolate, Hazelnut.

5. COFFEE & CHOCOLATE

- *Description:* Chocolate cake with Coffee chocolate whipped ganache.
- *Tags:* Chocolate, Coffee.

Category: Vanilla Based

1. CARAMALISED WHITE CHOCOLATE WITH ALMONDS

- *Description:* Vanilla cake with carmalised white chocolate ganache, almonds are optional.
- *Tags:* Caramalised white chocolate, Almond brittle, Vanilla.

2. PINA COLADA

- *Description:* Coconut & pineapple mousse, fresh pineapple.
- *Tags:* Pineapple, Coconut.

3. PINEAPPLE

- *Description:* Pineapple compote whipped ganache, vanilla cake.
- *Tags:* Pineapple, Vanilla.

Category: Tea Time

1. RICH MAWA

- *Description:* Indian inspired mawa cake made almond flour, all purpose flour, cardamom & butter.
- *Tags:* Mawa, Almond flour.

2. PERSIAN CAKE

- *Description:* Almond flour, orange juice, mawa, all purpose flour.
- *Tags:* Almond, Orange.

3. BUTTER CAKE

- *Description:* Butter, butter and lots of love. Classic butter cake.

Category: Seasonal

1. STRAWBERRY & CHOCOLATE

- *Description:* Strawberry compote | chocolate ganache | fresh strawberries.
- *Tags:* Strawberry, Chocolate.

2. STRAWBERRY & VANILLA

- *Description:* Vanilla cake, strawberry whipped ganache, fresh strawberries.
- *Tags:* Strawberry, Vanilla.

Part 3: Implementation Plan

This section outlines the chronological phases to transition the initial static designs into a functional, data-driven application with automated communication.

Phase 1: Flutter UI Development & Aesthetic Polish

- **Static UI & Prototyping:** Build the mobile screens using Flutter with mock data, establishing the core layout and luxury aesthetic.
- **Micro-Animations:** Implement subtle scaling, fade, and slide transitions when navigating the app.
- **Premium Imagery:** Integrate high-quality images placeholders to ensure the visual fidelity meets editorial standards.

Phase 2: Backend Data Modeling (Prisma & Supabase)

To accurately represent the patisserie domain, we will design the database schema to reflect the catalog above.

- **Categories:** "Chocolate Based", "Vanilla Based", "Tea Time", "Seasonal".
- **Products & Tags:** The exact item list, descriptions, and flavor tags mapped directly to the database.
- **Price Tiers:** Relational mapping for the 600g and 1050g variations.

- **Orders:** Robust schema modeling user carts, completed purchases, payment status, and delivery expectations.

Phase 3: API Layer Distribution (Next.js & tRPC)

- **Menu Retrieval:** Develop endpoints to fetch the hierarchical menu (Categories -> Products -> Price Tiers) and serve it to the mobile app.
- **Transaction Endpoints:** Securely handle payload submissions from the mobile app to create new order records in PostgreSQL.
- **Authentication Pipeline:** Implement JWT/Session handling via NextAuth.

Phase 4: Flutter API Integration & Connection

- **Data Injection:** Replace the temporary mock data with live dynamic data fetched from the Next.js API.
- **Checkout Flow:** Seamlessly transition users from the cart to the payment confirmation screen by connecting transaction endpoints.

Phase 5: WhatsApp Automation Integration

- **WhatsApp CTA:** Incorporate "Order via WhatsApp" or "Chat with Us" buttons in the app utilizing `url_launcher` to deep-link directly into a pre-populated WhatsApp chat with the `+91 9113231424` business number.
 - **Trigger Webhooks:** Set up webhooks inside the Next.js API that trigger whenever an order state changes.
 - **Message Templates:** Register approved WhatsApp templates for "Order Received", "Order Preparing", and "Out for Delivery".
 - **Customer Support Pipeline:** Integrate a chatbot flow using WhatsApp Business API that allows users to instantly request the menu or check order status over WhatsApp.
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